# Sidharrth Nagappan

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## Research Interests

Deep Learning, Natural Language Processing, Multimodal Learning, Low-resource Machine Learning

## Education

- ***Bachelor of Computer Science (Hons.) in Data Science*** Cyberjaya, MY
  *Multimedia University (MMU)* *Nov 2018 - Dec 2022 (Anticipated)*
  - **Cumulative GPA: 3.99/4.00**- Anticipated First Class Honours with Distinction (Summa Cum Laude)
  - Foundation Year GPA: 4.00/4.00 - Deep Learning Specialisation (deeplearning.ai)
  - Dean's list for every semester with a perfect 4.0 for 7 consecutive semesters
  - Distinction in Industrial Deep NLP thesis (Document Scoring Using Natural Language Processing) supervised by Ts. Dr. Goh Hui Ngo
  - **Community Lead @ Google Developer Student Clubs**: Organised hackathons, arranged talks and taught workshops on Python, Javascript, Machine Learning, Git and Flutter with a cumulative total of 1000 participants. Was actively involved in the Google Developer Space (GDG) within KL and the greater Malaysian community.
  - **President @ Hackerspace MMU**: Cyberjaya's largest hacker-maker group that promotes an engineering culture within the wider student developer community, in operation for over 10 years.
- ***Cambridge International General Certificate of Secondary Education (IGCSE)*** Cyberjaya, MY
  *elc International School* *July 2016 - July 2018*
  - **9A*s** for 9 subjects- Mathematics, Additional Mathematics, First Language English, Biology, Chemistry, Physics, History, Business Studies and Malay
  - Crystal 1st Rank- Graduated at the top of the class

## Publications

- **Sidharrth Nagappan***, Hui-Ngo Goh, Amy Hui-Lan Lim. Hybrid Deep Neural Networks for Industrial Text Scoring *In Proceedings of The 35th International Conference on Industrial, Engineering & Other Applications of Applied Intelligent Systems, Kitakyushu, Japan*, 2022.

## Experience

- **AirAsia** Kuala Lumpur, MY
  *Data Engineering Intern (DataOps and MLOps)* *May 2022 - August 2022*
  - The data engineering team within AirAsia Superapp builds and scales our DataOps and MLOps infrastructure, feeding production data into our lake for distributed analysis and predictive modelling.
  - Partnered with data analysts, data scientists, marketing and strategy to understand data and reporting requirements, adapt SQL and BigQuery scripts in Apache Airflow pipelines to keep up with changing business demands and implement data pipelines on distributed Kubernetes clusters for regular ETL ingestion.
  - Maintained lake by monitoring for structural irregularities and backfilling from transactional databases and SFTP servers on CentOS and GCP Kubernetes Engine.
  - Built and deployed dockerized automation scripts to federalise reporting across internal communication platforms for AirAsia's entire data infrastructure.
- **Sambal SOS (https://www.sambalsos.com)** Kuala Lumpur, MY
  *Co-Founder and Head of Engineering* *July 2021 - Present*
  - Malaysia's first open-source aid data crowdsourcing platform with over 30000 users, inspired by the pandemic. We went viral, scaled rapidly, and optimised the server after building the initial version in under 4 days.
  - Led a team of 5 developers across all ends of the stack, defined code practices, CI/CD and Docker workflows. Wrote frontend PWA using React.js (Chakra UI), backend server using Node.js (Express.js) and PostgreSQL, with image bucket on an AWS VPS. Spun up production clusters and scaled on GCP for thousands of simultaneous logins. Set up a custom PostgreSQL server on multiple EC2 and DO droplets with nginx and load balancers.

- Merged over 100 PRs from the Malaysian open source community. Was featured in over 20 major publications in Malaysia and spoke on multiple prestigious forums such as BFM and Malaysia's Big Data Week.

- **Fusionex Group (Research Partnership)** Kuala Lumpur, MY
  *Deep Learning Researcher in Natural Language Processing* *Aug 2021 - Nov 2021*
  - An industrial research project that aims to automate the adherence checking of annual reports through a novel hybrid framework, encompassing LSTMs, word-level keyword attention and linear attention transformers (Longformer and BigBird), using Pytorch and Tensorflow, with extended research into computational long content scoring for the processing of >4000-word reports.
  - Collated the world's first labelled corpus of financial text from the Malaysian corporate ecosystem.
  - Trained high-performance text scoring models on GCP Compute Clusters, with deployment as microservices using FastAPI and Docker.
  - Defined a new avenue for multimodal natural language processing, with the introduction of the first framework to apply dual, word-level attention in tandem with neural transformers, and the first deep learning framework for the processing of text in non-academic and industrial settings.

- **Rodeo** Kuala Lumpur, MY
  *Senior Staff Software Engineer (Contract)* *Jan 2022 - Present*
  - Rodeo provides transit advertising solutions, and the company needed a new app that connects potential clients to marketing campaigns. As of Series B, the company has a valuation of RM 20 million.
  - The two-man engineering team at Rodeo is rebuilding the MVP from the ground up, including a real-time mobile app for drivers, an admin interface, a gamified app for the general public and a super app server.
  - The super app's server is scaled across Dockerized GCP compute engines, with MongoDB clusters and GCS image buckets. Direct credit payments are automated through TnG digital's API and Stripe, with payment and email automation modules.
  - Optical Character Recognition (OCR) and Deep Image Segmentation models are hosted on GCP Pub/Sub for automated identity verification.

- **Lava X** Puchong, MY
  *Software Engineer (Part Time)* *June 2021 - Aug 2021*
  - Built an MNC e-commerce CMS platform using Typescript, React (Next.js), Nest.js and GraphQL

- **Auronex** Cyberjaya, MY
  *Software Engineer (Part Time)* *Oct 2020 - Nov 2020*
  - Integration between the frontend Flutter client and backend API for a fitness application.
  - Worked on a Node.js RESTful API and a React frontend for an accounting platform and made custom modifications to multiple themed components for clients.

## Notable Projects

- **Vending.js** : PWA that automatically routes technicians to nearest appointments based on current and past locations for efficient dispatch | *React.js, Node.js (Express.js), MongoDB*
- **Common Sans** : Credit profile analysis engine powered by web scrapers and NLP using social media and e-commerce review data with support for Manglish sentiment and time-series activity - *React.js, Tensorflow, Flask, Selenium*
- **Semantic Style Transfer** : Trained and deployed pipeline that combines fast style transfer with UNet semantic segmentation for aesthetically stunning blended reproductions | *Tensorflow, Pytorch, Streamlit*

## Technical Skills

- Languages and frameworks: Python (Tensorflow, Keras, Pytorch, Scikitlearn, Pandas, Flask, OpenCV, Airflow), Javascript (Typescript, React.js, Redux, MobX, Next.js, React Native, Node.js, Express.js, Nest.js, Apollo), Dart (Flutter), Java (JavaFX), C++, HTML/CSS, PHP, MongoDB, PostgreSQL, MySQL, Redis
- CI/CD and Tools: GCP, Docker, Kubernetes

## Awards

- **RHBHack 2021**: Special Challenge Winner and Runner-up
- **AtlasHack 2021**: Special Challenge Winner and Runner-up
- **The Dean's List**: Faculty of Computing and Informatics MMU
- **Crystal 1st Rank**: IGCSE Topper
- **KUMON Mathematics Completer**: Finished KUMON Mathematics at the age of 13, ranking 6th in the country among my age group.
- Duke of Edinburgh International Award (DoFE Bronze)
- **Model United Nations (MUN)**: Finished as an outstanding delegate, chairperson and director at various MUN conferences in the Malaysian scene
- **Merit Scholar**: Merit scholarships throughout pre-university and undegraduate years

## Speaking Engagements

- **BFM.my**: Building Sambal SOS
- **Malaysia's Big Data Week 2021**: Spoke among key Malaysian figures about using tech for community good
- **The Center of Applied Data Science**: Malaysian Tech Changemakers
- **Bawatana.co**: Non-profit startups
- **The Star, Malaysiakini, Free Malaysia Today, Says.com, Varnam**: Interviews about Sambal SOS
- **Google Developer Student Club @ University Utara Malaysia**: Getting started in tech and software engineering